TEXORA · SESSIONAL NOTES · PRACTICAL STUDY SHEET

Practical Study of Cotton Spinning Machines: Comber, Speed Frame, Ring Frame, Doubling and Reeling

A working guide to the five machines of the combed-cotton line: what to identify at each machine, which settings to read and record, and the production points your lab report must carry — with the comber head and the ring frame yarn path drawn for tracing on the spot.

TE-2202 · SPINNING TECHNOLOGY-II SESSIONAL · STUDY 1 · LEVEL 2 TERM 2

1. Object and method of the study

Object: to connect the theory of TE-2201 with the machines themselves — to identify every working part of the comber, the speed frame (simplex), the ring frame, the doubling frame and the reeling machine, to read their settings as the mill has set them, and to record the production figures each setting produces.

Method — the same routine at every machine. Work in this order and your report writes itself:

1. **Trace the material path** from feed package to delivery package, naming every element the material touches. If you can walk the path without hesitation, you understand the machine.
2. **Identify the working parts** listed in each section below, and state in one line what each part does.
3. **Read and record the settings** — gearing-driven values (speeds, draft, twist) and gauge settings (distances, penetrations). Ask the lab instructor before touching any setting; settings are observed, never altered.
4. **Record production points** — the quantities in the worked examples: feed weight, delivery weight, speed, efficiency, waste percentage.
5. **Sketch** the material path (Figures 1 and 2 are your models) and note faults seen in operation — end breaks, waste accumulation, damaged parts — with their likely causes from the theory sheets.

2. The comber

2.1 Parts to identify

- **Lap stand and lap roll** — the prepared lap (from the lap former, TE-2201 Topic 1) unwinding at the back of each head.
- **Feed roller** — the fluted roller that advances the lap by the set feed length each cycle.
- **Nipper assembly** — lower nipper plate and upper nipper knife on the swinging frame; the clamp that holds the fringe.
- **Combing cylinder (half-lap)** — the shaft clothed over part of its circumference with needle or saw-tooth segments; carries the **index wheel** (40 divisions) on its shaft end, from which all timings are read.
- **Top comb** — the single needle blade that drops into the fringe during detachment only.
- **Brush roller and suction** — strips the cylinder of noil and carries it away.
- **Detaching rollers** — the pair that grips the combed fringe, reverses for piecing-up, and draws the web off.
- **Web table, trumpet, draw box and coiler** — condense the eight webs, draft them together and coil the sliver into the delivery can.

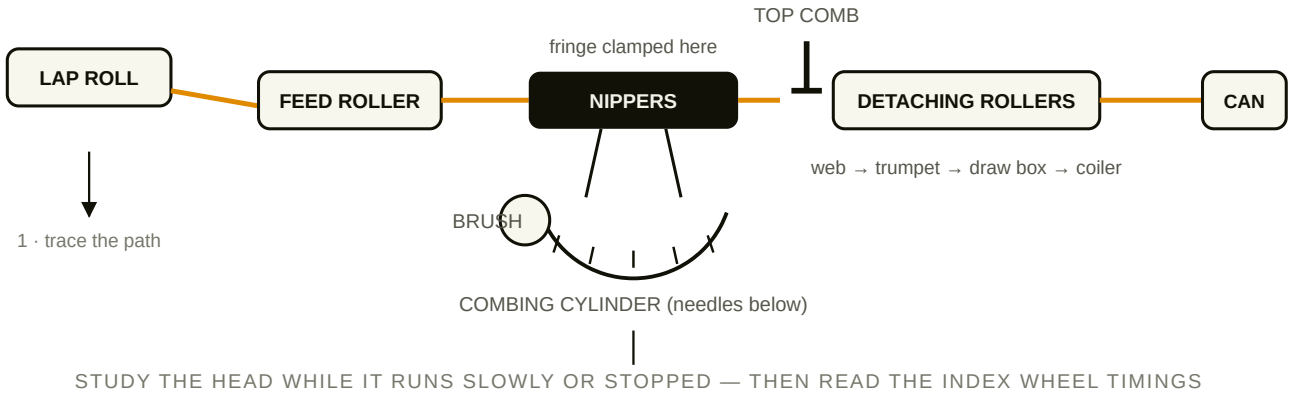


Figure 1 — Study schematic of one combing head. Follow the lap: roll → feed roller → nippers → cylinder (which combs the fringe from beneath the nipper swing) → detaching rollers, with the top comb entering only at detachment; the web then passes to trumpet, draw box and can.

2.2 Settings and observations to record

ITEM	WHAT TO WRITE IN THE REPORT
Number of heads; lap weight	Heads working (usually 8); lap weight in g/m, measured by weighing a 1 m length
Feed length and feeding type	mm per nip; forward or backward feeding (TE-2201 Topic 2), and the index-wheel position at which feeding occurs
Cylinder speed	Nips per minute, counted or from the machine display

Detachment setting	Distance in mm between nipper bite and detaching-roller nip at closest approach
Top comb setting	Penetration depth as set (relative to the fringe plane)
Noil percentage	Collect the noil over a timed run (or weigh feed lap against delivered sliver) and compute waste %
Delivered sliver	Sliver weight (g/m or hank) and the draw-box draft across the combined webs

PRODUCTION CHECK — COMBER

Sliver production (kg/h) = feed per nip (g) × nips/min × heads × 60 × efficiency × (1 − noil fraction) ÷ 1000, where feed per nip = feed length (m) × lap weight (g/m). Measure the four machine values during the study and verify the machine's actual output against this figure.

3. The speed frame (simplex)

3.1 Parts to identify

- **Creel** — holds the comber (or draw-frame) sliver cans; slivers are guided over skewer rails to the drafting rollers.
- **Drafting system** — on the lab machine, find whether it is roller-only or a **3-over-3 apron system**; locate back, middle (apron) and front roller pairs and the condensers between zones.
- **Flyer and spindle** — the flyer rotates about the bobbin and guides the roving to the winding point; find the flyer legs, the presser finger, and the spindle that carries the bobbin.
- **Cone drums and differential** — trace the drive from the main shaft: one cone drum drives the bobbin through the **differential (epicyclic) motion**; the second input changes as the builder motion shifts the belt along the cones.
- **Builder mechanism** — the ratchet-and-pawl and reversing arrangement that shortens the bobbin-rail traverse layer by layer to form the tapered ends of the roving bobbin.

3.2 Settings and production points to record

- **Total draft and break draft** — from the change wheels or the displayed gearing; state which zone carries the main draft.
- **Roving hank** (Ne) of the delivered roving, by weighing a measured length.
- **Twist** — roving twist in turns per inch, from flyer speed ÷ front-roller delivery; note the false-twist effect of the flyer top where relevant.

- **Winding mode** — whether the machine is set for flyer lead or bobbin lead, and how you can tell from the drive.
- **Production** — front-roller delivery (m/min) and spindle speed (rpm). Production per spindle per shift (kg) = delivery × 60 × shift hours × efficiency × 0.4536 ÷ (768.1 × roving hank Ne), since one hank at count Ne weighs out over 840 × Ne yards (= 768.1 × Ne metres) per pound. Work it for the lab machine and compare with its rated output.

4. The ring frame

4.1 Parts to identify

- **Creel and roving bobbin** — the speed-frame package feeding each spindle.
- **Drafting system** — three roller pairs with double aprons in the main zone; locate top rollers (cots), the aprons, and the weighting arm that loads the rollers.
- **Spindle** — blade, wharve (driven by tape from the tin roller) and bolster; the cop (yarn package on its tube) sits on the blade.
- **Ring and traveller** — the flange ring fixed in the ring rail; the C-shaped traveller clipped on the flange, through which the yarn passes.
- **Lappet guide, balloon control ring and separator** — the guide centres the yarn over the spindle; the yarn whips around between lappet and traveller as the **balloon**.
- **Builder motion and ring rail** — the cam-driven rail that rises and falls, shortening its stroke to build the cop's tapered chase.

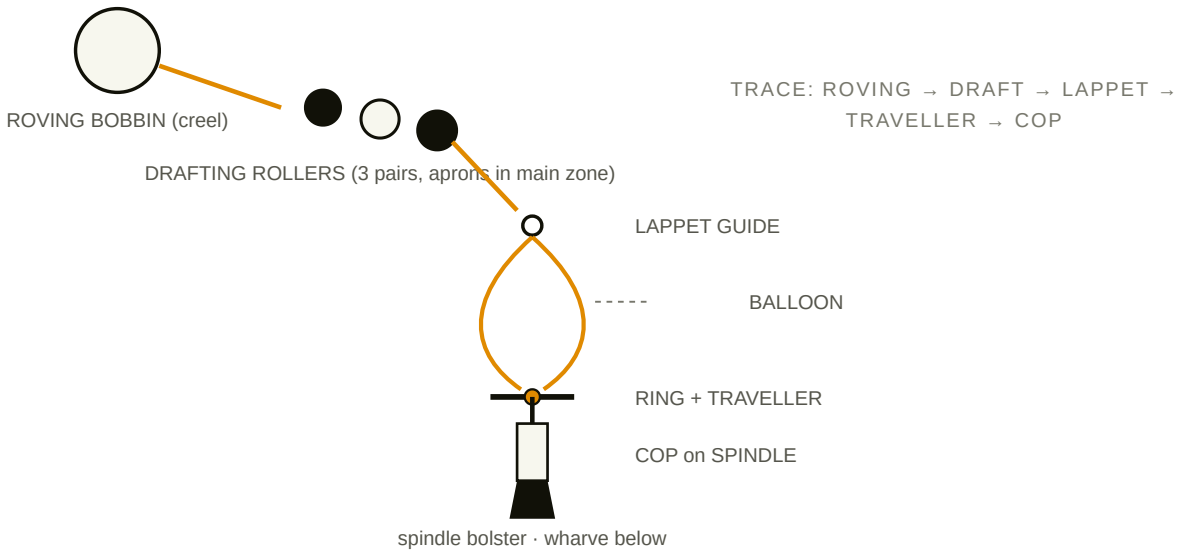


Figure 2 — The yarn path at one ring-frame spindle. The front rollers deliver the drafted strand; the lappet centres it over the spindle; centrifugal force balloons it out to the traveller, whose lag behind the spindle both inserts twist and winds the cop.

4.2 Settings and production points to record

- **Spindle speed** (rpm) and **front-roller delivery** (m/min); from these, **twist** = spindle speed ÷ delivery (turns per unit length, converted to tpi) — cross-check with the twist change wheel fitted.
- **Total draft** and the **yarn count** being spun (Ne), verified by reeling a lea (Section 6).
- **Ring diameter** (mm) and **traveller number/type** fitted; note balloon height and whether balloon control rings are in use.
- **End breakage** — count breaks at a group of spindles over a timed observation and express as breaks per 100 spindle-hours, the mill's standard form.
- **Production per spindle** (kg) = delivery (m/min) × 60 × hours × efficiency × 0.4536 ÷ (768.1 × count Ne) — the same hank relation as at the speed frame. Record also the doffing time and the cop content (g per cop).

5. Doubling and reeling — the finishing studies

5.1 Doubling

Parts to identify: the supply packages (two or more single-yarn cops or cheeses), the doubling winder's guides, tension device and winding drum that first combine the ends into one package, and — where the lab has one — the **doubling (ring) frame** that twists the plied yarn.

- **Observe and record:** number of plies; whether doubling is dry or wet (trough) doubling; the twist direction of the singles (usually Z) and of the ply (usually S, chosen to balance the singles' twist); the resultant count — two 40s singles give a 2/40s yarn of resultant count ≈ 20s.
- **Record** the tension arrangement and any clearing device, and note why even tension between ends matters (a slack end loops and corkscrews around the tight one).

5.2 Reeling

Parts to identify: the reel (swift) of standard circumference **1.5 yd (54 in)**, the yarn guides, the rapid traverse, and the counting/knocking-off motion that stops the reel at the set length.

- **Observe and record:** 80 revolutions × 1.5 yd = **120 yd = 1 lea (skein)**; 7 leas = **1 hank = 840 yd** — the standard cotton hank, on which the count system Ne is based (count = number of 840-yd hanks per pound).
- **Record** how a lea is weighed for count testing: count = (leas × 840 yd) relation applied as Ne = length (yd) ÷ (840 × weight in lb) for the sample weighed.
- **Note** the tying of leas into hanks and the bundling/baling that follows — reeling is the bridge between spinning and every count test and sale form that follows.

6. What the lab report must contain

SECTION	CONTENT
Machine identity	Maker, model, year if shown on the plate; number of heads/spindles; the material and count being processed during the study
Material path	Your traced path with a labelled sketch (Figures 1–2 as models)
Parts table	Each working part, its location, and its function in one line
Settings record	The settings tables of Sections 2–5, filled with the observed values — never copied from the theory sheet
Production check	One worked production calculation per machine from your own observed figures
Faults observed	Any fault seen in operation, its probable cause from TE-2201, and the remedy the theory suggests

7. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. On the comber, which single setting most directly fixes the noil percentage, and how would you verify the noil actually being extracted during the study?
2. The index wheel has 40 divisions. What is it for, and which three events of the combing cycle are timed against it?
3. A speed frame is found running with bobbin lead. What does that mean, and where in the drive does the speed difference come from?
4. Using Figure 2, explain how one device — the traveller — performs two jobs at the ring frame.
5. Two cops of 40s single yarn are doubled and twisted together. State the resultant count designation and the twist direction you would expect, with the reason.
6. A lea reeled from the ring frame weighs 0.955 g… show how the count is obtained from a lea weighing, taking 1 lea = 120 yd and 1 lb = 453.6 g. What count does 0.955 g give?

Answer key

1. The **detachment setting** — the gap between nipper bite and detaching-roller nip; only fibres long enough to bridge it are drawn off, so it sets the short-fibre cut-off. Verification: collect and weigh the noil over a timed run and divide by the lap fed in the same time (or compare feed-lap weight with delivered sliver weight).

2. It is the timing reference of the whole cycle: one cylinder revolution = one cycle, and the 40 divisions let the mechanic set and check the instants of feeding, nipper closing/opening, top comb entry and detaching-roller reversal relative to cylinder position.
3. Bobbin lead means the bobbin rotates slightly faster than the flyer, and the yarn is wound on because the bobbin pulls it around the flyer leg. The difference is produced in the differential motion: a constant main-shaft drive plus a varying cone-drum drive are combined so the bobbin's speed margin changes with package diameter as the builder shifts the cone belt.
4. The yarn passes through the traveller, which is dragged around the ring by the spindle-driven yarn but lags because of ring friction and air drag. The spindle's full rotation twists the yarn between front rollers and traveller; the lag — spindle revolutions minus traveller revolutions — is exactly the number of coils laid on the cop in the same time. One rotating difference does the twisting, the lag does the winding.
5. Designation 2/40s, resultant count ≈ 20s (the plied yarn is twice as heavy per unit length as one 40s end, ignoring twist contraction). Singles are normally Z-twist; the ply is twisted S so that the plying twist works against and balances the singles' twist, giving a stable, round, non-lively yarn.
6. Count Ne = length (yd) ÷ [840 × weight (lb)]. Weight = 0.955 ÷ 453.6 = 0.002105 lb. Ne = 120 ÷ (840 × 0.002105) = 120 ÷ 1.768 ≈ **67.9, about 68s**. (Check against the nominal count on the machine: a lea weighing is the laboratory's verification that the frame is spinning what its gearing claims.)
